

Fig. 6: Internet connection



Fig. 7: Star connection



Fig. 8: Ring connection



Fig. 9: Bus connection



Fig. 10: Tree connection



Fig. 11: Mesh connection

by connecting the different LANs using services such as dedicated leased phone lines, dial-up phone lines, satellite links and data packet carrier services.

Internet. The Internet is a global system of linked networks that are worldwide in scope. It serves millions of users with data communication services like World Wide Web, e-mail, file transfers, IPTV and much more. It uses the standard Internet protocol suite (TCP/IP). Internet connection is shown in Fig. 6.

Intranet. Intranet is a network employed within a business organisation to offer various network services to its employees. It offers ease of sharing and connectivity between the employees.

Examples include Outlook Service and Lync Communicator tool.

Network topologies

A network topology is how computers, printers and other devices are connected over a network. It describes the layout of wires, devices and routing paths.

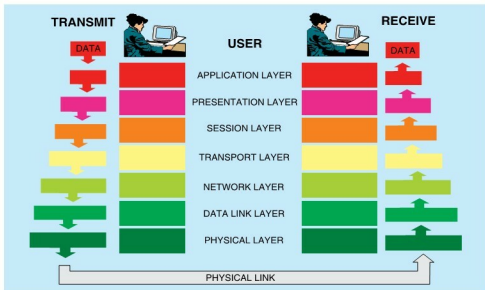


Fig. 12: OSI model

Some common network topologies include star, ring, line, bus and tree configurations.

Star. In this topology, one central node is connected to each of the other nodes on a network—similar to a hub connected to spokes in a wheel. This central connection allows a fully functioning network even when other devices fail.

Ring. In this topology, each node is connected to two other nodes in its immediate vicinity, forming a ring. It can be visualised as a circular configuration that requires at least three nodes.

Bus. In bus topology, each node is connected to a central bus that runs along the entire network. All the information transmitted across the bus can be received by any system in the network.

Tree. In tree topology, one 'root' node connects to other nodes, which, in turn, connect to other nodes, forming a tree structure. Information from the root node may have to pass through other nodes to reach the end node. If the device on top of the chain fails, the entire network will be down.

Mesh. Full-mesh topology connects all the nodes together. It creates the most redundant and reliable network around, especially for large networks. If any link fails, there is always another link to send data through.

The OSI layer model

OSI is an internationally recognised, non-proprietary standard for network-

ing and operating system involved in network functions. It is a fundamental model to all the communications between network devices. The OSI model (see Fig. 12) consists of seven layers:

Application layer. At the application layer, a Web browser serves as the user interface for accessing websites. This layer is closest to the user. Specifically, it consists of HTTP interfaces between the Web browser and the Web server.

Presentation layer. This layer pertains to the semantics and syntax of the information and thus ensures that the information sent by the application layer of one system is readable by the application layer of the other system. Also, it provides features of data encryption and compression.

Session layer. The session layer establishes a connection between the client and the server. It enables two network resources to hold ongoing communications across a network.

Transport layer. This layer ensures reliable delivery of data from the Web server to the client. Layer 4 protocols include TCP and UDP.

Network layer. This layer involves logical addressing of the sender and the receiver. Routers are used to ensure that the best path is chosen for data flow.

Data layer. At this layer IP addresses are translated into hardware addresses.